

Sample test

QUESTION 1 [4 marks]

Solve for x .

i) $2 \ln x = \ln(2x - 1)$.

ii) $1 < e^{2x-2} < 2$.

QUESTION 2 [3 marks]

Find the exact value of

$$\sin \left(2 \arccos \frac{3}{5} \right).$$

QUESTION 3 [1 mark]

The displacement (in meters) of a particle at time t (in seconds) is given by $y(t) = t^2 + 4t$. Find the average velocity of the particle on the interval $[1, 2]$.